Testing and Validation of Automated Road Vehicles

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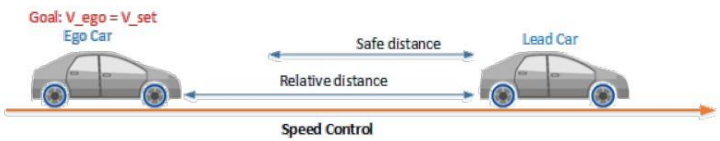
Project objective

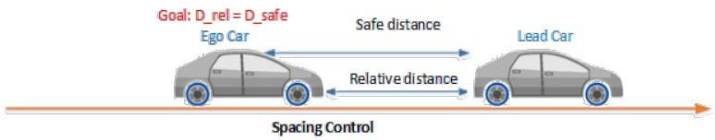
- **The Goal:** To evaluate the safety, comfort, and efficiency of a standard ACC controller under stochastic driving conditions
- **The Approach:**
  - Model the system
  - Define nominal parameters
  - Define Input factors and their ranges
  - Latin Hypercube Sampling (LHS) for efficient parameter space coverage.
  - Running the simulations and collecting data
  - Analyzing the results

System Under Test (ACC)

- **ACC System (our case)**
  - Driver set a desired speed v_{des}
  - Desired acceleration u(t) computed on
    - longitudinal speed of the Ego-Vehicle; relative distance to the ahead leader vehicle; relative speed to an ahead leader vehicle.
  - Two operating mode
    - If there is no ahead vehicle, or if $\Delta d(t) \geq d_{safe}(t)$, the goal is to reach and maintain the desired-set speed (speed control mode).
    - If $\Delta d(t) < d_{safe}(t)$ the goal is to slow down and maintain the safe distance (spacing control mode).
  - $d_{safe}(t) = d_{st} + v_{ego}(t) \cdot h$
    - h is the minimum safe time-gap distance
    - d_{st} is the distance at standstill
  - $\Delta d(t)$ = actual relative distance w.r.t. vehicle ahead

$$u(t) = f(x) = \begin{cases} \min\{K_v e_v(t); [K_p e_p(t) + K_w \Delta v(t)]\}, & if\ e_d \leq 0 \\ K_w \Delta v(t) - K_p e_d(t), & if\ e_d > 0 \end{cases}$$





- The ego vehicle travel at a driver-set velocity while maintaining a safe distance from the lead car.

Input Variables

Vehicle specific inputs	vehicle mass
	Frontal area
	Drag coefficient
	Ego maximum deceleration
System specific inputs	Time gap
	Set speed
Environment specific inputs	Slope of the road

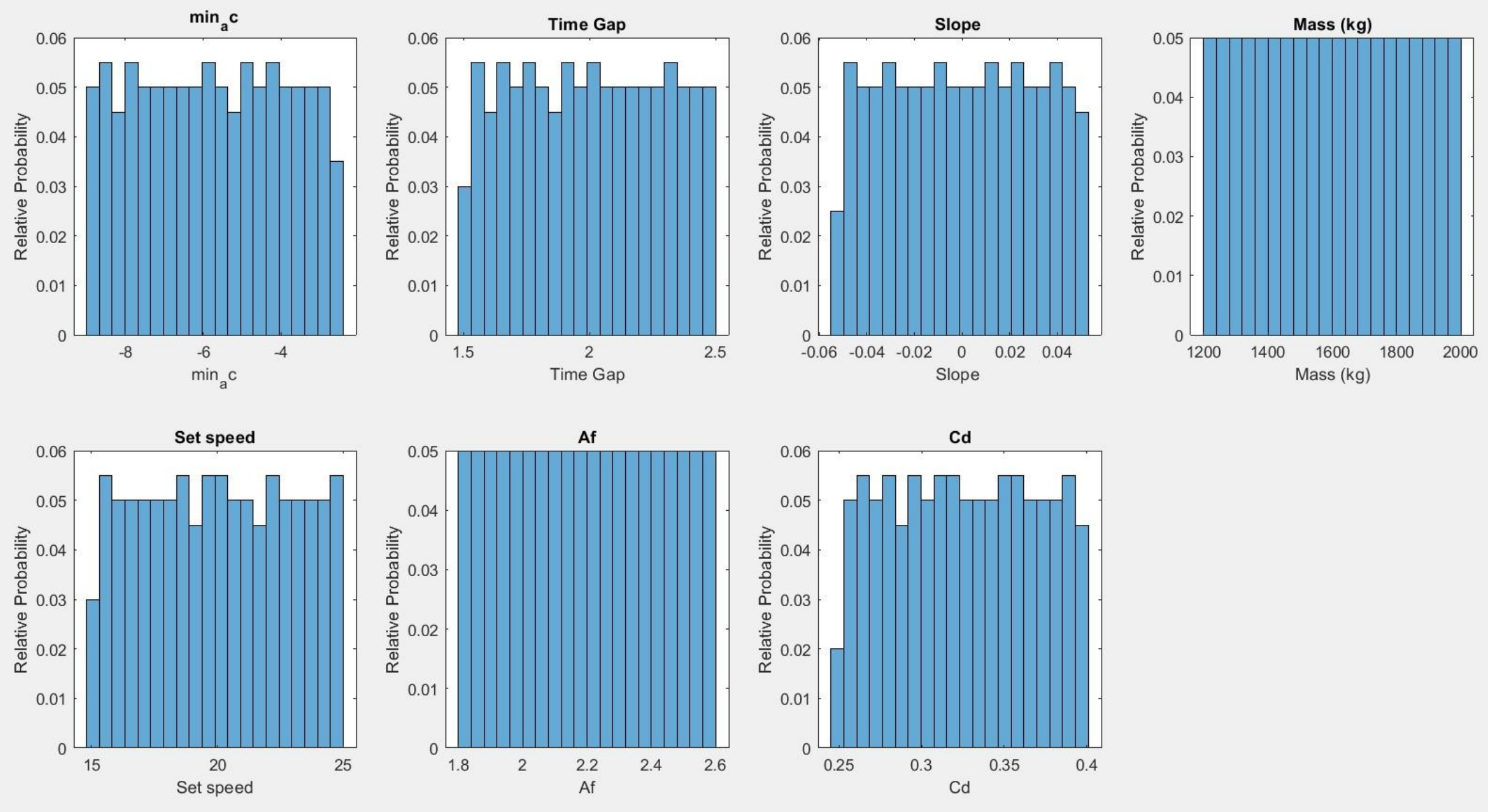
Input Sampling Validation

- Two checks
  1. Distribution check : compared the actual mean and variance with the theoretical target. As shown in the table, at N = 200, the error is very low.
  2. Independence check : verified that the inputs are not correlated to each other.
- The tests show samples following a uniform distribution with no correlation between the inputs

SAMPLING QUALITY CHECK (LHS N=200)

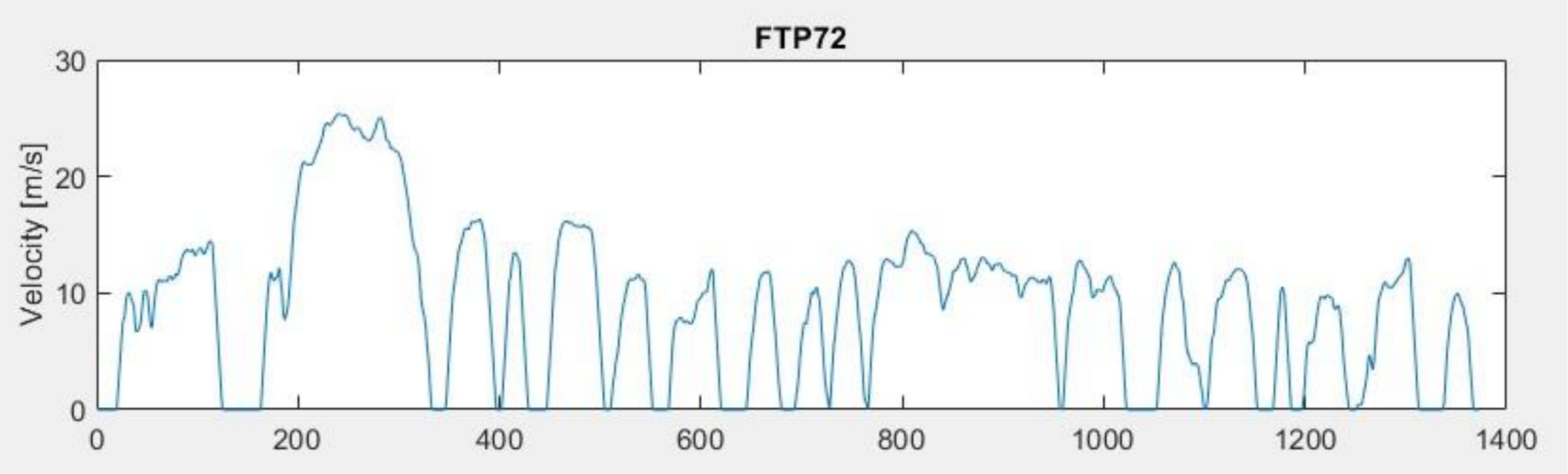
Variable	Th. Mean	Act. Mean	Err %	Th. Var	Act. Var	Err %
min_ac	-5.7500	-5.7499	0.00%	3.5208	3.5390	0.52%
Time Gap	2.0000	2.0000	0.00%	0.0833	0.0838	0.50%
Slope (deg)	0.0000	0.0000	0.00(Abs)	0.0009	0.0009	0.75%
Mass (kg)	1600.0000	1600.0677	0.00%	53333.3333	53617.7208	0.53%
Set speed	20.0000	19.9999	0.00%	8.3333	8.3785	0.54%
Af	2.2000	2.2000	0.00%	0.0533	0.0536	0.54%
Cd	0.3250	0.3250	0.00%	0.0019	0.0019	0.60%

Average Correlation between Inputs: 0.0586 (Target: 0.0000)


min_a c
Relative Probability
0.06
0.05
0.04
0.03
0.02
0.01
0
-8
-6
-4
min_a c
Time Gap
Relative Probability
1.5
2
2.5
Time Gap
Slope
Relative Probability
-0.06
-0.04
-0.02
0
0.02
0.04
Slope
Mass (kg)
Relative Probability
1200
1400
1600
1800
2000
Mass (kg)
Set speed
Relative Probability
15
20
25
Set speed
Af
Relative Probability
1.8
2
2.2
2.4
2.6
Af
Cd
Relative Probability
0.25
0.3
0.35
0.4
Cd

Leader Drive scenario

- The FTP72 drive cycle is used as the source for the leader vehicle velocity



KPIs – Safety

- Minimum Time To Collision
  - the time until a crash between two objects if both vehicles were to continue on their current path at their current velocities.

- $\text{TTC} = \frac{(X_L - X_F)}{v_F - v_L}$

- TET, time exposed to time to collision
  - total accumulated duration (in seconds) during a trip where the Time-to-Collision is below a specific safety limit
  - Acquired by integrating TTCI over time
- TTCI
  - a measurement of whether the defined threshold is exceeded.

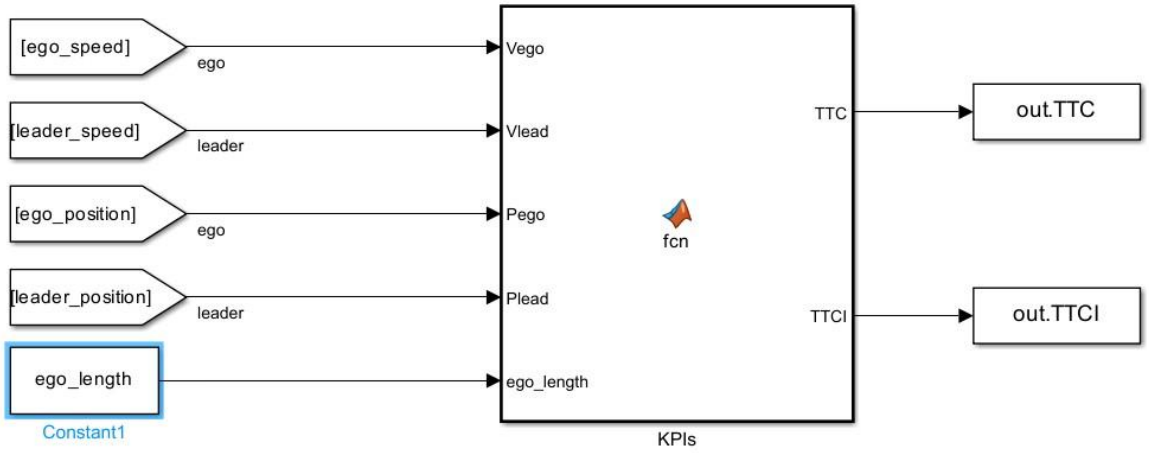
- $\text{TTCI} = \begin{cases} 1, & if\ TTC(t) < Threshold \\ 0, & otherwise \end{cases}$

KPIs – Safety - Implementation

KPIs

```
function [TTC,TTCI]= fcn(Vego, Vlead, Pego, Plead, ego_length)

delta_v = Vego-Vlead;
delta_p = Plead-Pego-ego_length;

% calculating Time to collision
if (delta_v > 0 && delta_p > 0)
%the follower car is catching up to the leader and they didn't collide yet
    TTC = delta_p/delta_v;
elseif delta_p == 0 % a collision has happened
    TTC = 0 ;
else
    TTC = 1000; %We are safe
end

%capping the value of TTC at 1000
if TTC > 1000
    TTC = 1000;
end

%calculating TTCI

TTCI = TTC< 3;
```

```
%KPIs for safety

%minimum time to collision for each run
Output_Matrix(i,1) = min(ans.TTC);

%TET, amount of time when the ttc was under the safety threshold
Output_Matrix(i,2) = sum(dt*ans.TTCI);
```



KPIs – Comfort and Energy consumption

- RMS jerk
  - RMS jerk quantifies the continuous oscillation and roughness of the vehicle's acceleration over time, serving as a metric for passenger ride comfort and motion sickness.
- KWH per 100 km
  - efficiency metric that normalizes total energy usage over distance

```
% KPI for comfort
jerk = [0; diff(ans.Ego_Acceleration) / dt];
RMS_Jerk = sqrt(mean(jerk.^2));
Output_Matrix(i,3) = RMS_Jerk;


%KPI for Energy consumption per 100 kilometer
P_inst = ans.Instantaneous_power; % instantaneous power = Force*velocity
P_inst(P_inst<0) = 0; %filter out negative energy
Total_Energy = sum(P_inst * dt)/ 3.6e6; %KWH, Integrate power over time to get energy
Total_distance = max(ans.Ego_Position)/1000; %KM
Output_Matrix(i,4) = (Total_Energy/Total_distance)*100; %KWH per 100 KM
```

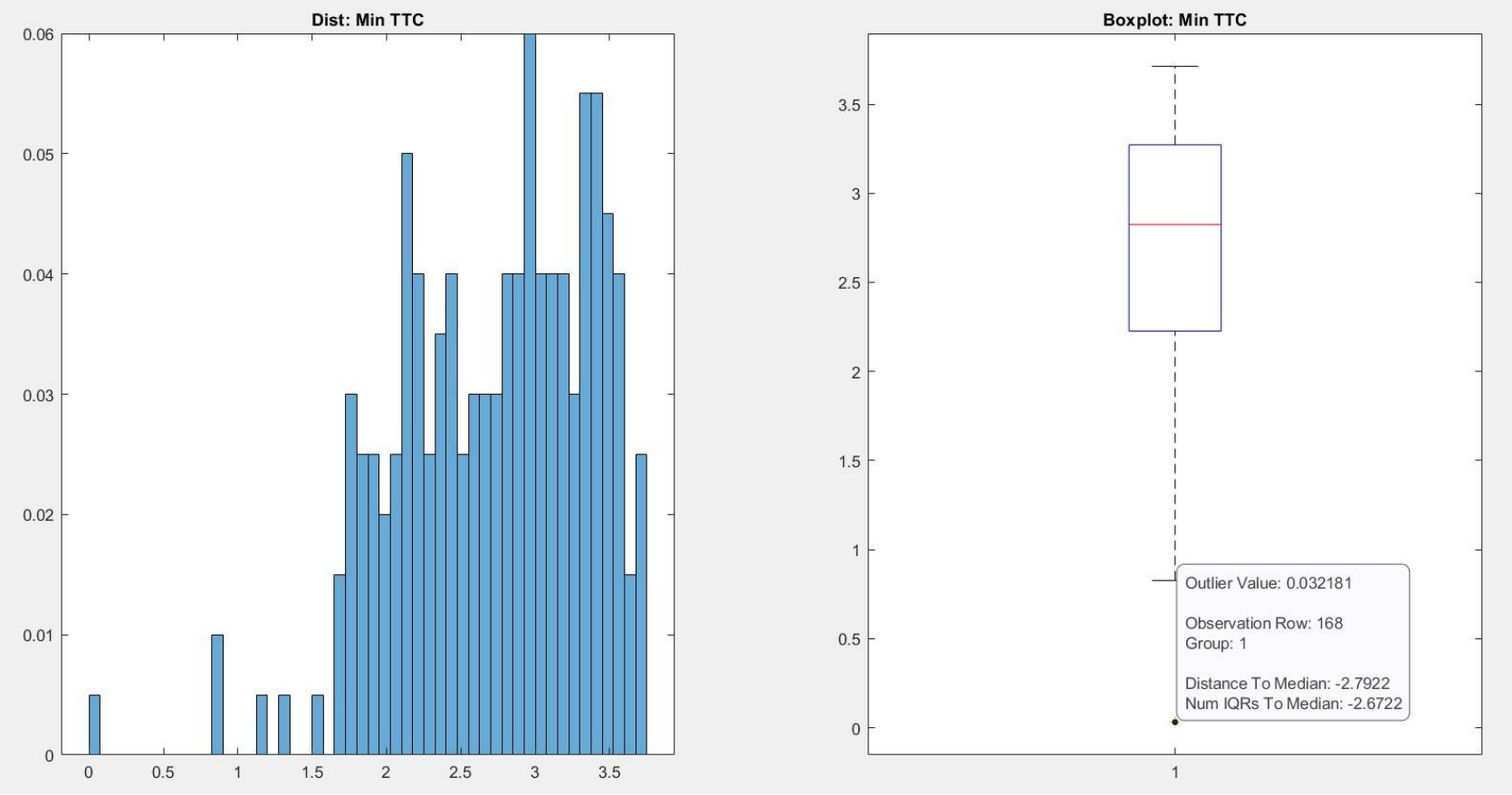
KPIs- System requirements

- RMS tracking error
  - **Definition:** The Root Mean Square difference between the *Safe Distance* (desired gap) and the *Actual Relative Distance*.
  - **Significance:** Quantifies the controller's ability to strictly obey the time-gap policy. A high error indicates either a "lazy" response (falling behind) or a "dangerous" response (tailgating).

- RMS Relative velocity
  - **Definition:** A measure of the speed difference between the Ego vehicle and the Leader. Ideally, this should be zero during steady-state following.
  - **Significance:** Represents the "damping" of the system. High values indicate a nervous controller that constantly accelerates and brakes to hunt for the target speed.

- RMS Set speed error
  - **Definition:** The deviation between the driver’s Selected Speed and the vehicle's Actual Speed
  - **Significance:** It quantifies how much the external traffic environment forces the driver to deviate from their desired travel plan.

KPIs - Safety – Min TTC - Analysis



From 200 runs, we have one outlier where the Min TTC is 0.0321s

KPIs - Safety – Min TTC - Analysis

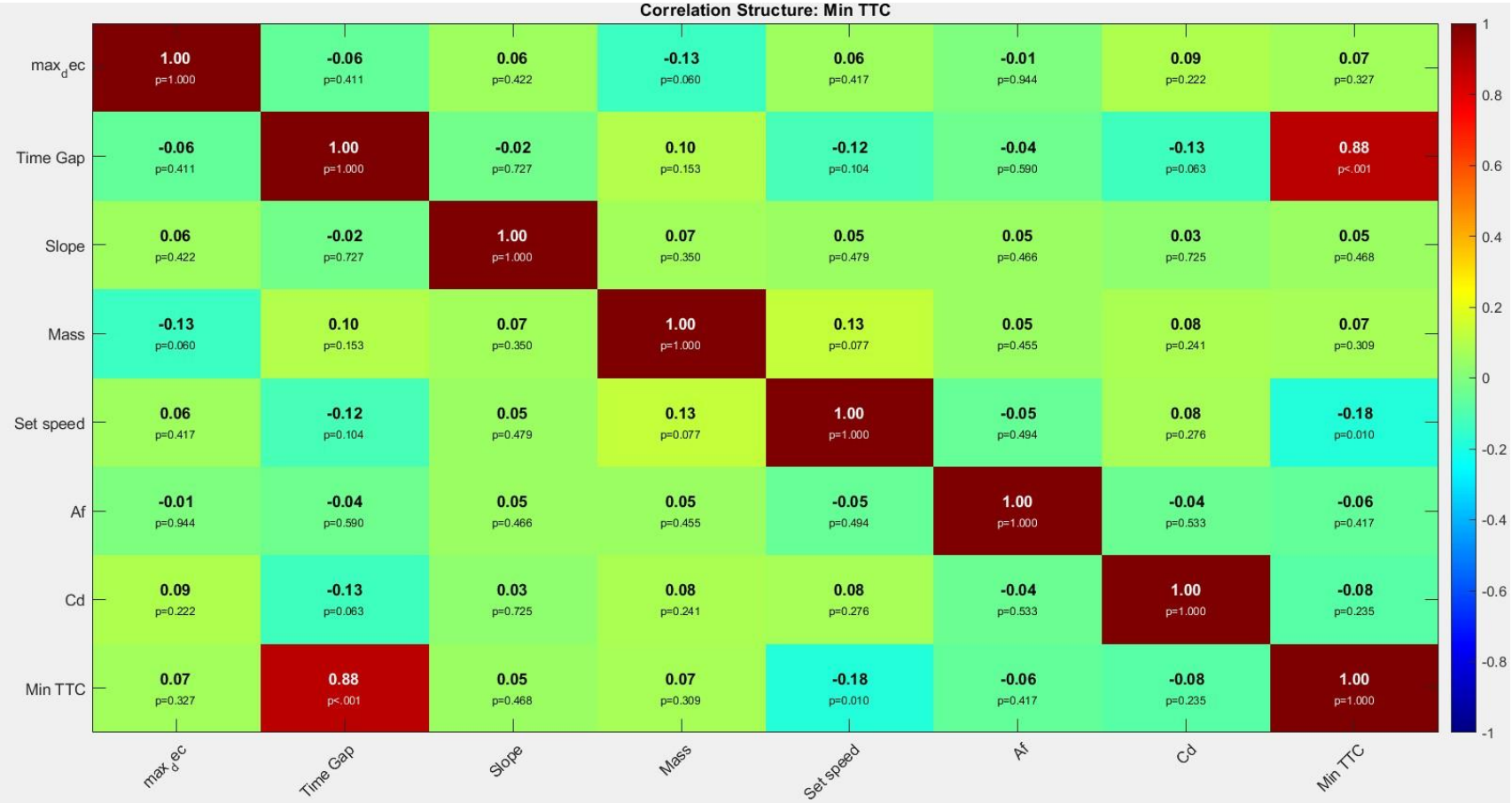
- **Run 168:** This specific concrete scenario with Max_Decel and Time_Gap were low within the sampled range combined with a decline angle caused the minimum time-to-collision to be very small

CRITICAL FAILURE ANALYSIS (N=1)

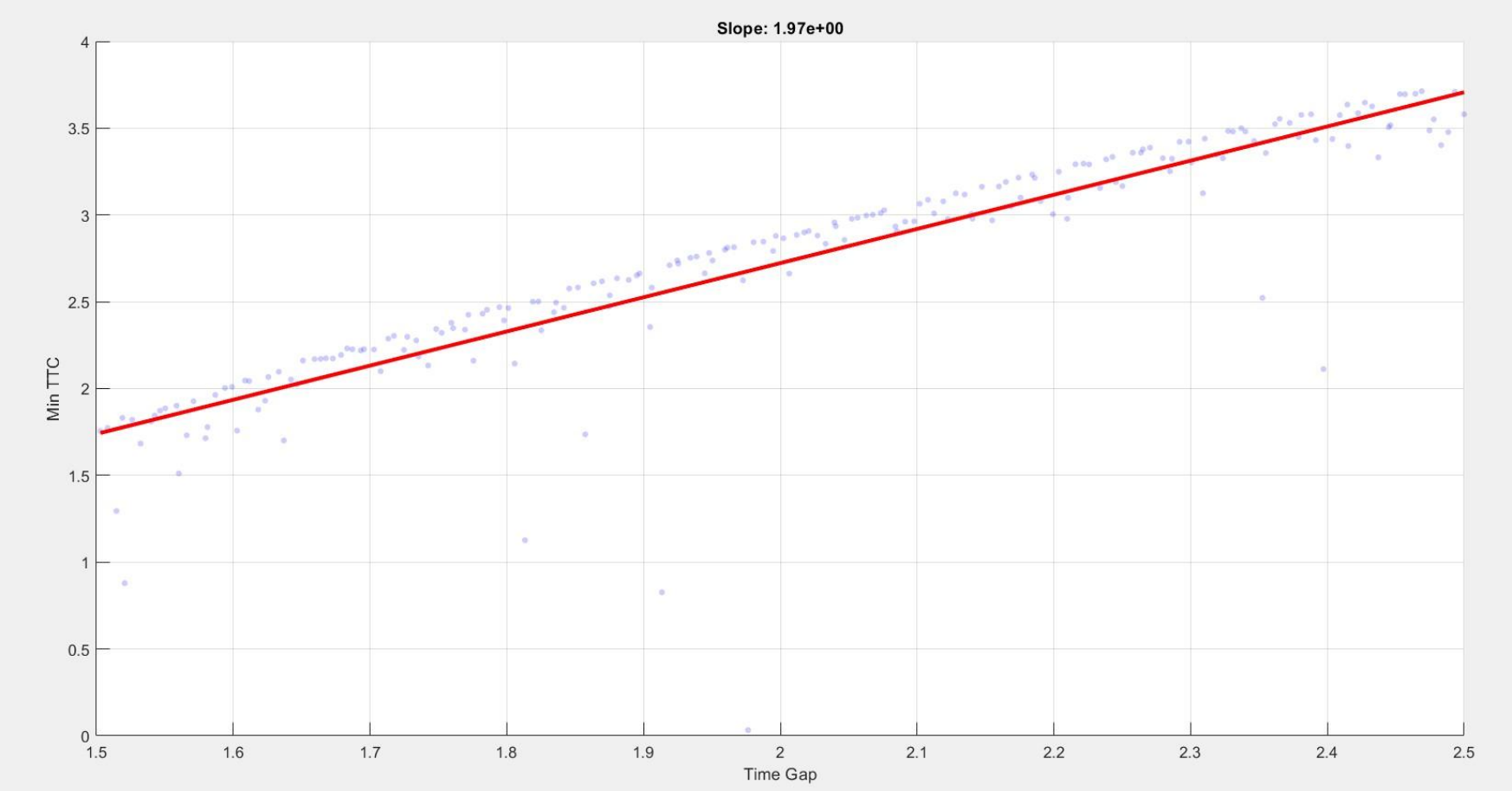
Run_ID	Max_Decel	Time_Gap	Slope_Deg	Mass_kg	V_Set_kmh	Af	Cd	Min_TTC
168	2.5629	1.9763	-2.5857	1300.6	81.098	2.0976	0.25905	0.032181

KPIs - Safety – Min TTC - Analysis

- **Dominant Factor: Driver Setting (Time Gap)**
  - **Correlation (R=0.88):** The Time Gap parameter is the most critical determinant of safety.
  - **Implication:** Safety is primarily defined by the user's choice of the controller parameters.

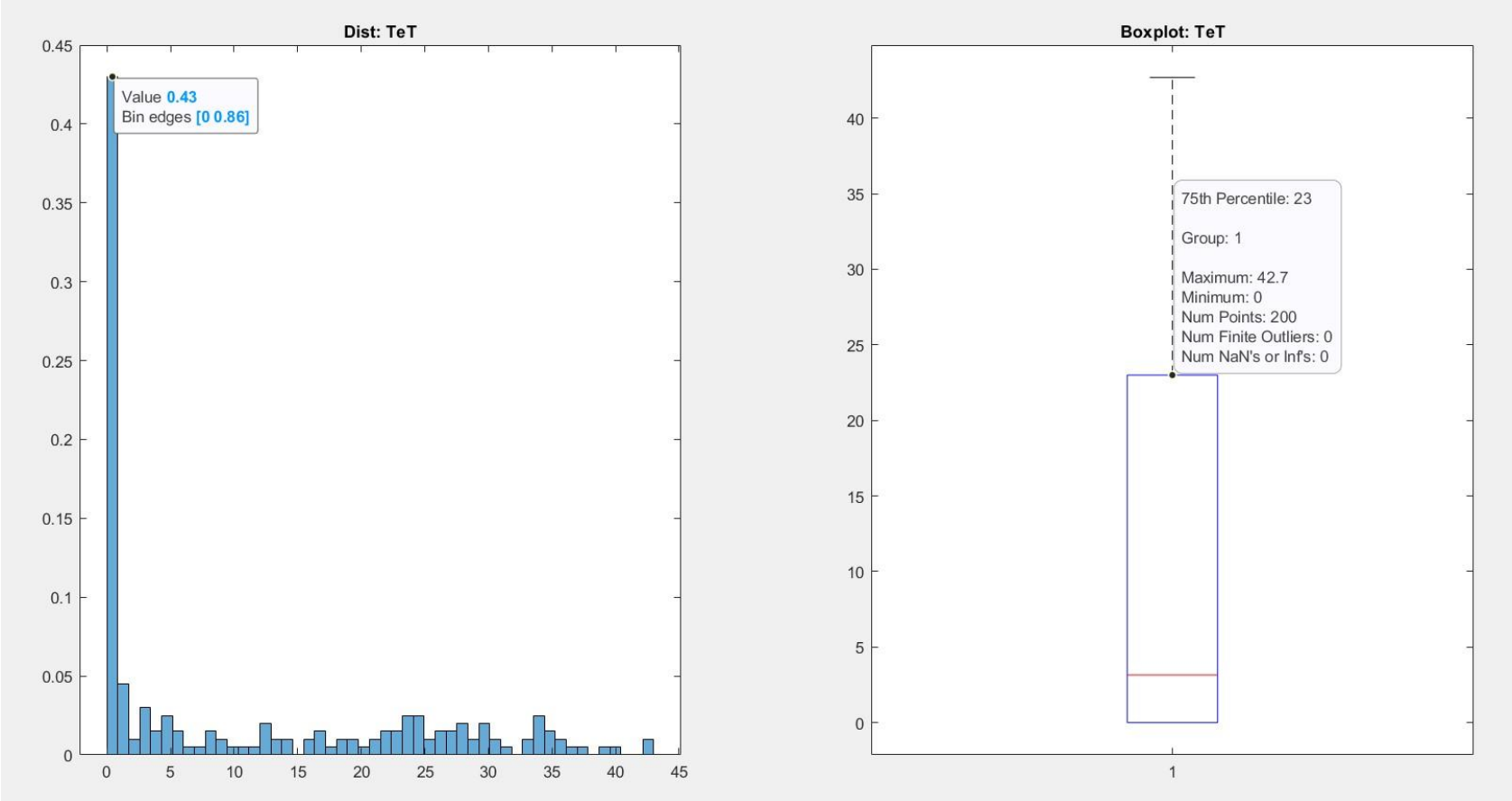


KPIs - Safety – Min TTC - Analysis



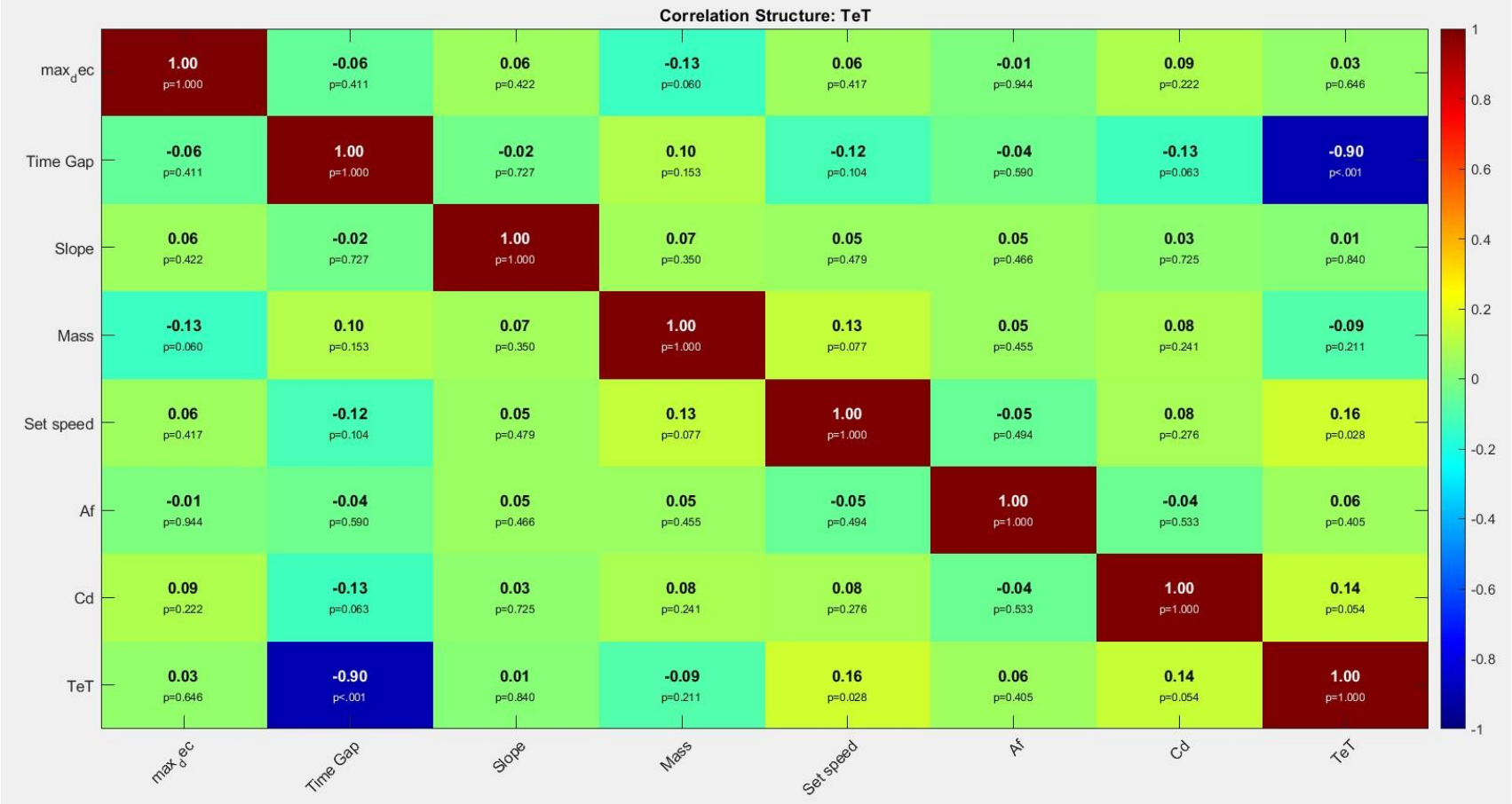
KPIs - Safety – TET-Analysis

- From 200 runs, TET was zero for 38.5% of them which explains the high value for the bin [0 0.86].
- The Boxplot show no outliers



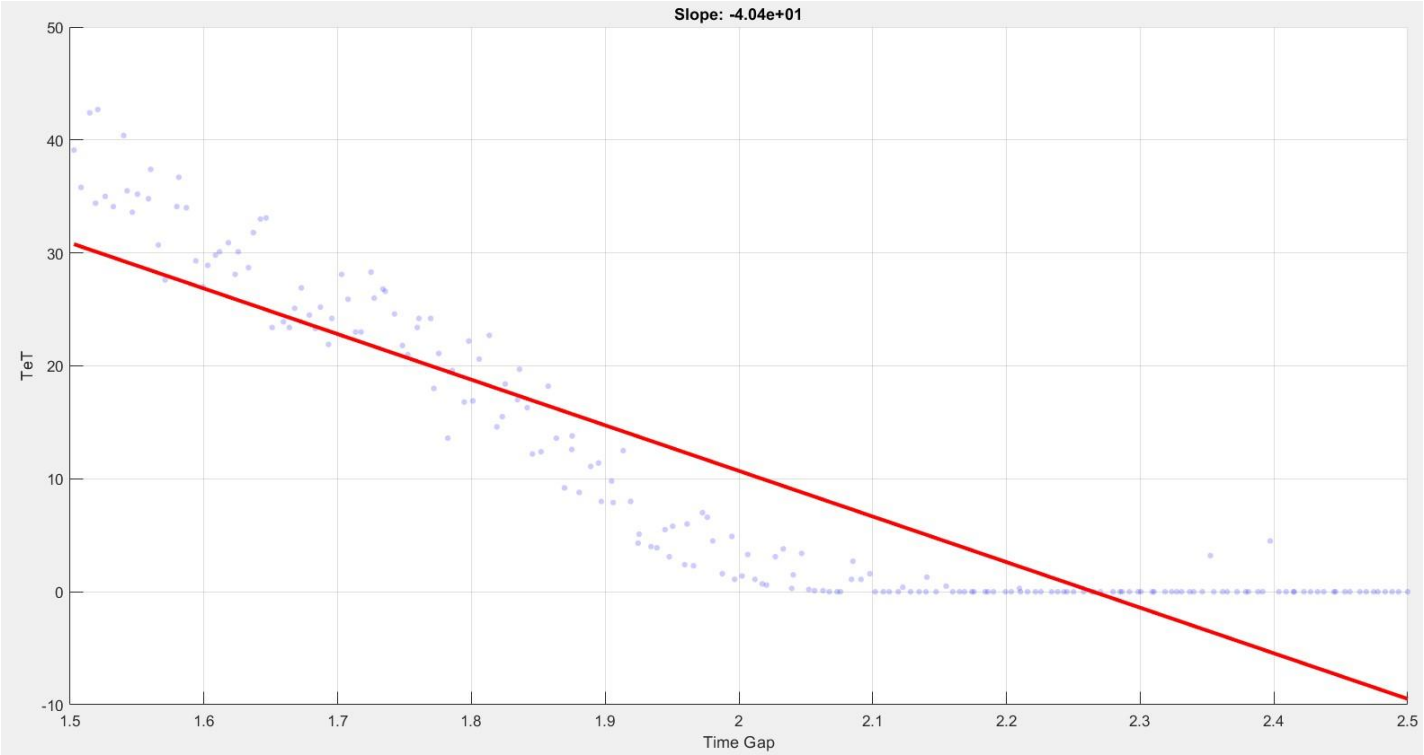
KPIs – Safety -TET– Analysis

- **Dominant Factor: Driver Setting (Time Gap)**
  - **Correlation (R=-0.9033):** The Time Gap parameter is the most critical determinant of safety.
  - **Implication:** Increasing the time gap results in decreasing the time a vehicle exposed to danger.



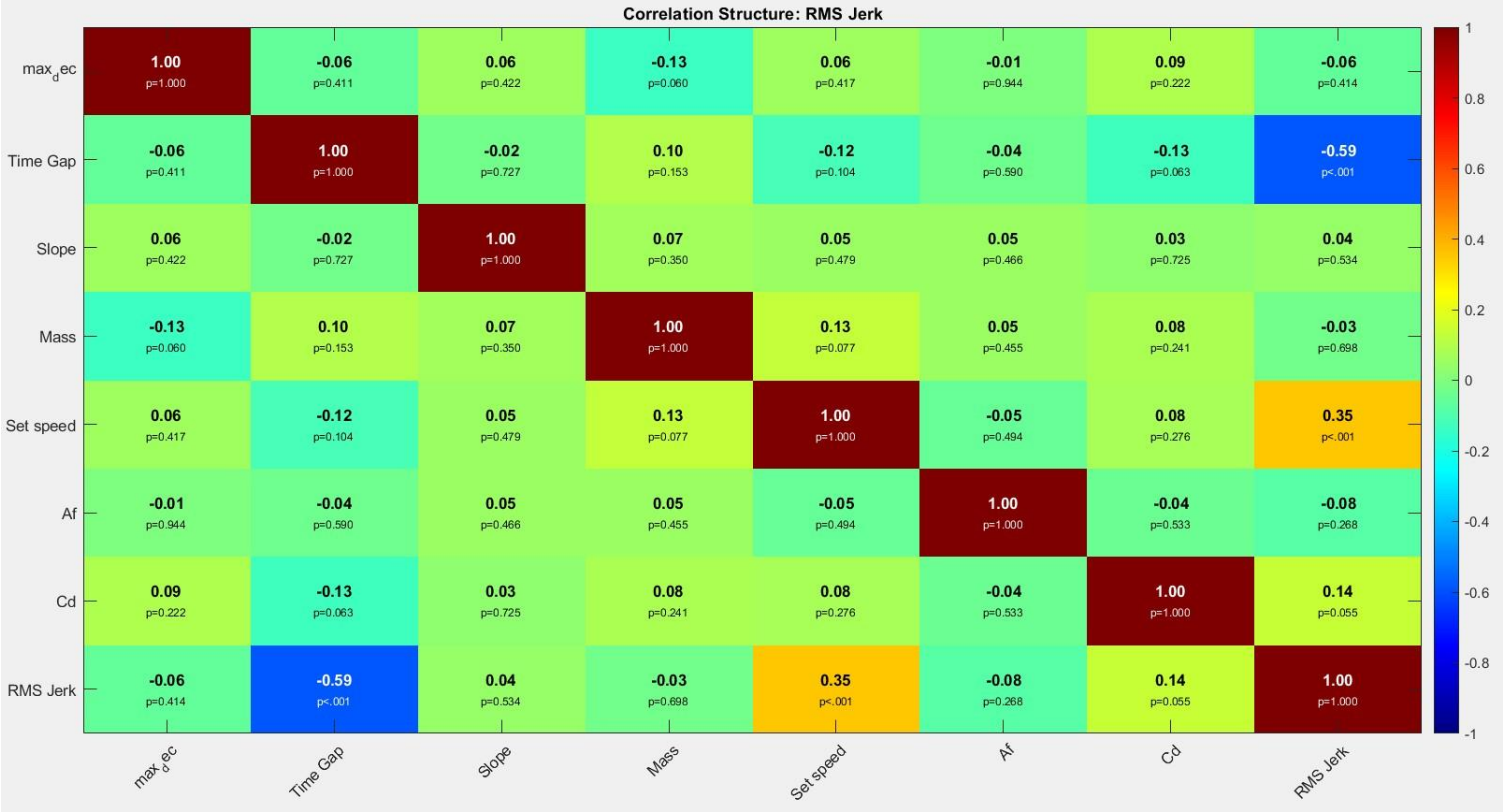
KPIs – Safety –TET-Analysis

The linear model is an approximation; TET is non-negative by definition.

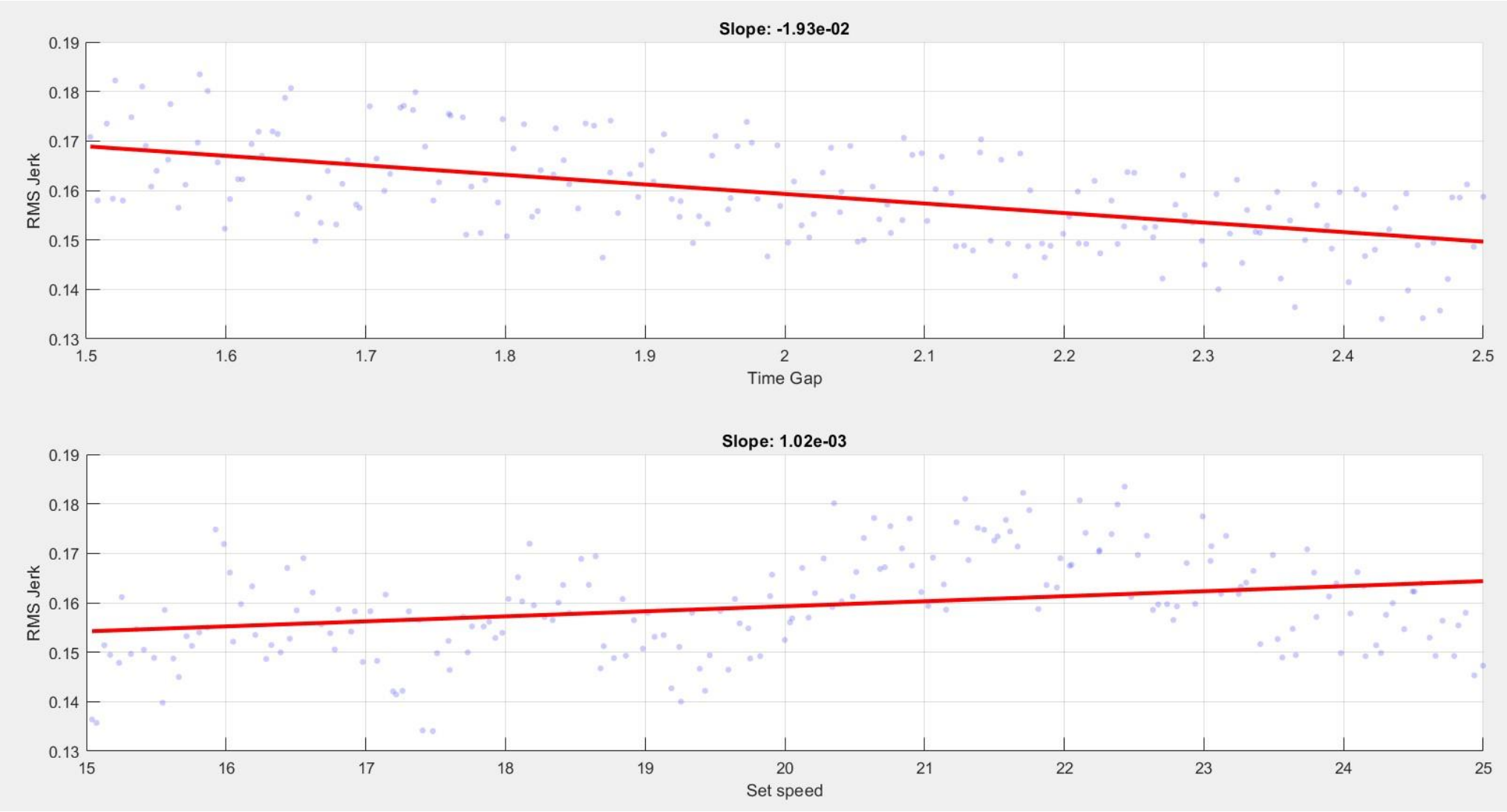


KPIs – RMS jerk

- Comfort is determined by two factors
  - A moderate negative correlation of **-0.59** exists between **Time Gap** and **RMS jerk**
    - Increasing the Time Gap creates a larger spatial buffer. This allows the controller to "filter out" the leader's noise and React with smooth accelerations.
  - A low positive correlation of **0.3544** exists between **Set Speed** and **RMS jerk.**
    - Increasing the set speed causes the controller to command more aggressive accelerations to reach the target speed. These sharper transitions lead to higher acceleration gradients.



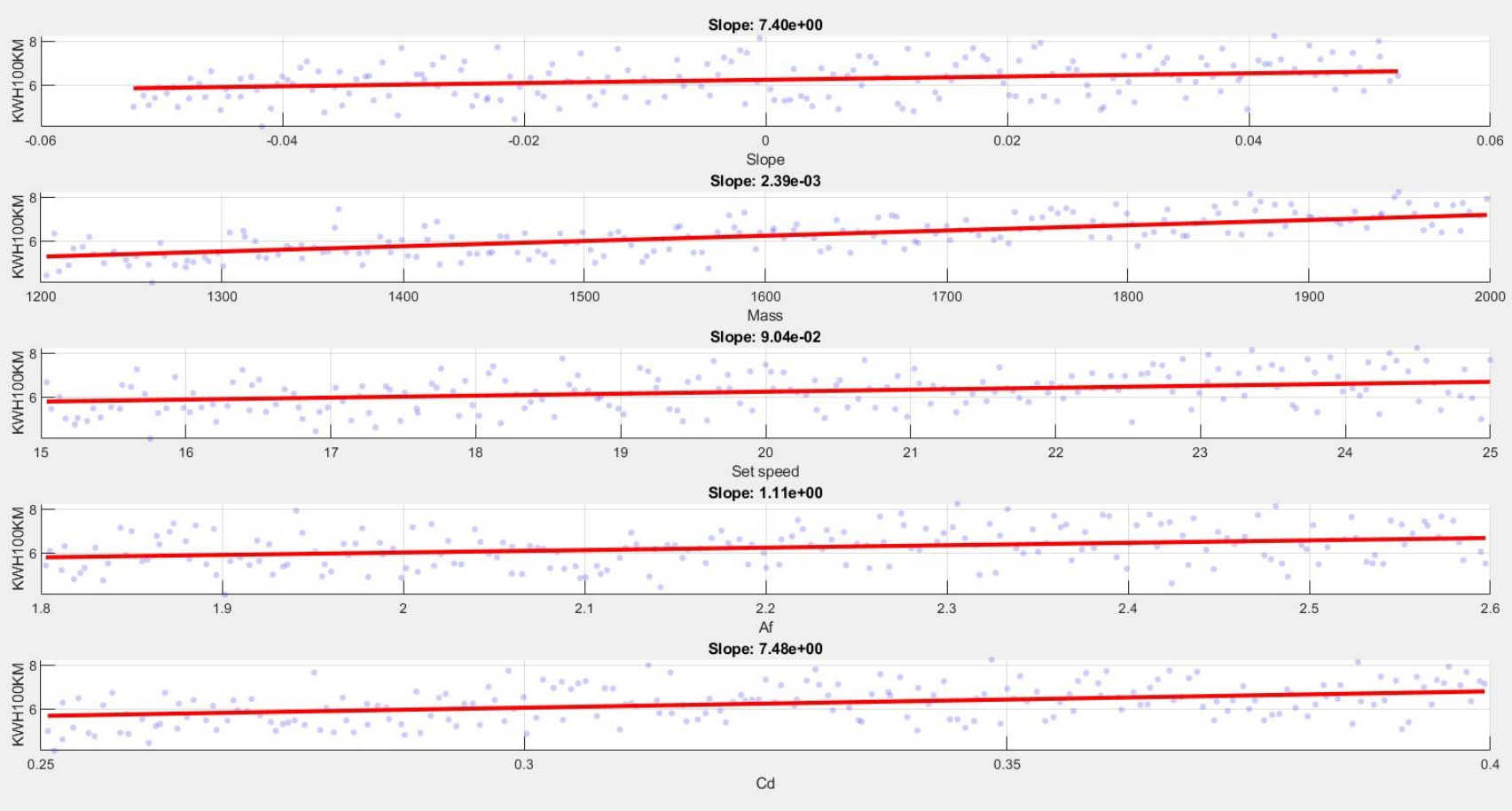
KPIs – RMS jerk



KPIs – KWH per 100 KM

- Energy Efficiency is determined by the *Physical Plant* (Mass, Aero), Set Speed and Slope.

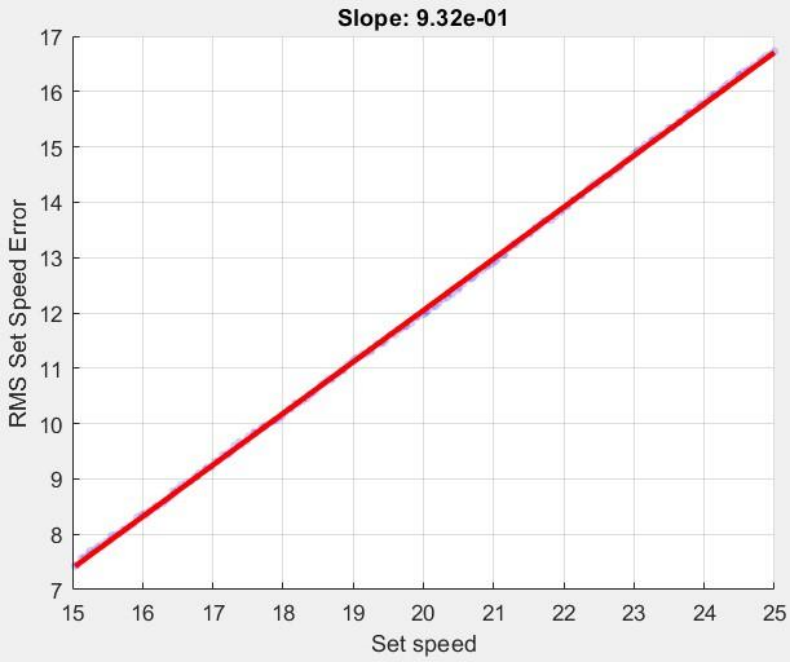
KPIs – KWH per 100 KM

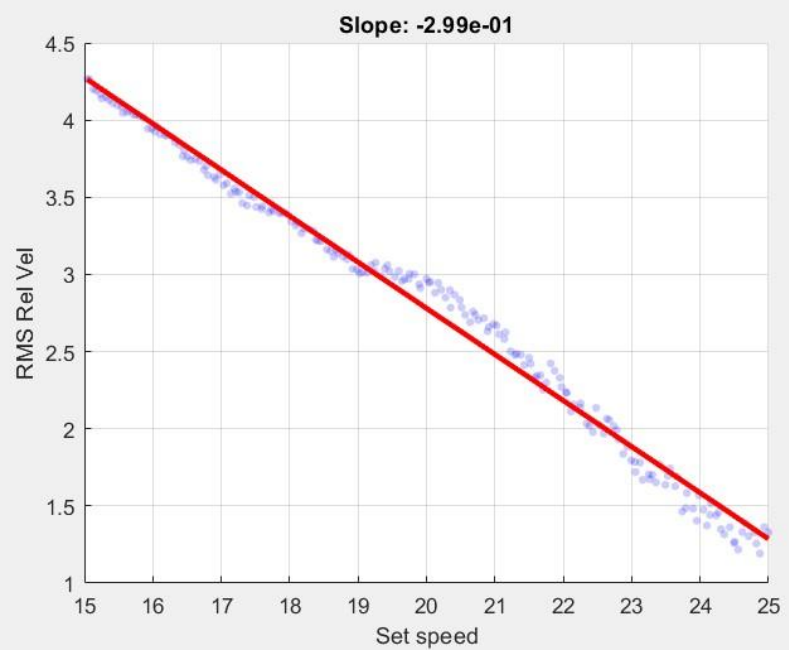


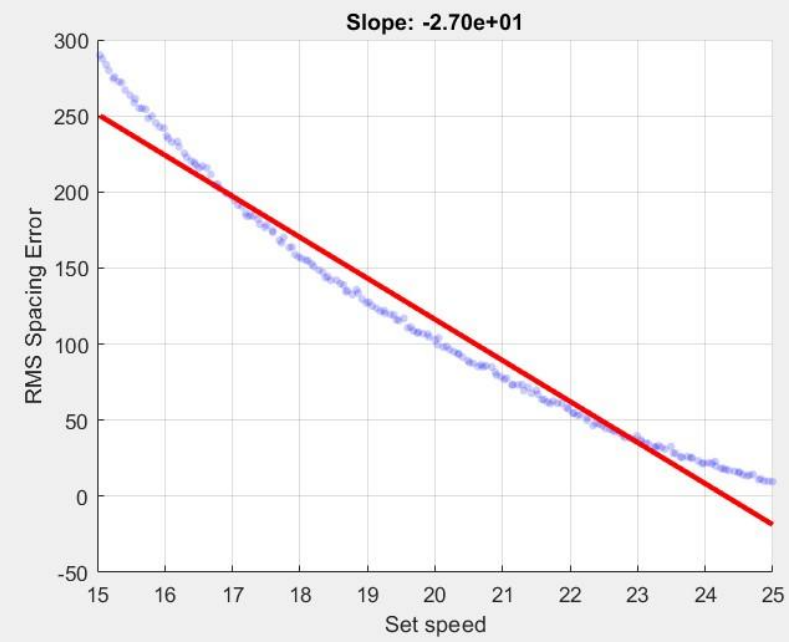
KPIs – controller requirements

- Spacing error and relative velocity both have very high negative correlation with set speed
  - Tracking errors decrease as Speed Limit increases
  - When the Set Speed is low, the controller lacks the authority to keep up with a fast leader.
  - The vehicle physically falls behind, creating large artificial tracking errors.
- Set speed error has a very high positive correlation with set speed.
  - The higher the set speed, the more often the vehicle is obstructed by the traffic.
  - This error quantifies the difference between what the driver wants and what the traffic allows.

KPIs – controller requirements







KPIs – controller requirements

The errors in KPIs #5 and #6 are due to the **maximum speed constraint** imposed by the driver. The leader is faster than the ego vehicle's set speed.

The error in KPI #7 is due to traffic conditions when the user sets a desired speed higher than the leader's speed.

The ACC gains are well tuned; *99.5% of runs (199/200) maintained Min TTC > 0.1s", and "38.5% of runs had zero safety exposure time*. The errors stem from user-defined or traffic constraints .